
Decodable but Not Actionable: Localizing and Releasing the Familiarity Throttle in Language Models

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Abstract

Language models linearly encode whether they are familiar with an entity, yet fabricate about it anyway. We give a mechanistic account of why the gap persists: the signal is throttled—read by a small, identifiable circuit at less than behavioral magnitude. On a five-stratum exposure ladder with per-entity verification, the internal signal tracks the volume of training coverage rather than existence: products announced during training but not released until after it read as familiar, released but sparsely covered products read as unfamiliar, and the model answers 74% of questions about verified never-seen real entities, where every answer is a fabrication. The familiarity signal transfers across unrelated content domains (0.95 AUROC, replicated on a second model family), while a contextual grounding signal transfers across corpora only weakly (0.59). Causally, the signal reaches the abstention decision through a two-stage circuit—entity-reading attention heads feeding decision MLPs, concentrated at layers 15–22, five of 1,188 components closing 80% of the gap—operating below behavioral magnitude. Conditioning on the signal releases it: a frozen steering vector gated by the probe recovers calibrated abstention (0.905 flagging at the untouched model’s own 0.015 false-refusal rate) where the same vector always-on and output-confidence fine-tunes collapse to indiscriminate refusal.

1 Introduction

A line of interpretability work shows that language models linearly represent whether an entity is known (Ferrando et al., 2025; Obeso et al., 2025; Orgad et al., 2025), and that this representation is causal for hallucination and refusal and present in the base model (Ferrando et al., 2025; Lindsey et al., 2025); concurrent work argues that verbalized miscalibration is a “readout failure” rather than a knowledge deficit (Miao & Ungar, 2026). Taking the existence of the signal as given, we ask three further questions. *What* does the signal encode—that an entity is fictional, or that the model is unfamiliar with it? *How* does it generalize—is it a universal circuit or a distribution-specific artifact? And *where*, causally, does it fail to reach behavior, and what does that imply about how to make the model act on it? Throughout, the action at issue is the abstention decision: flagging non-knowledge rather than committing an answer. Recovering correct content is outside its scope; steering cannot provide it, and an unfamiliar entity precludes it.

Our contribution, on a question-answering testbed with constructible ground truth (matched real and fictional supply-chain entities, with transfer validated on unrelated domains), is a mechanistic account of the throttle, building on the established existence and base-model origin of entity-knowledge features (Ferrando et al., 2025; Lindsey et al., 2025):

- **What the signal encodes.** A five-stratum exposure ladder, verified entity-by-entity at construction, shows the signal encodes graded familiarity: products that never existed during training read as familiar when their announcements were covered, and long-released but sparsely covered products read as unfamiliar (Section 4).
- **Opposite generalization signatures.** The familiarity signal is a domain-general structure of the pretrained model (cross-domain AUROC 0.954), while a construct-distinct grounding signal

is corpus-specific (0.59 across corpora), identifying the poles of the probe-transfer spectrum with distinct epistemic constructs (Section 5).

- **Localization.** The signal’s causal effect on abstention concentrates at layers 15–22 under a norm-matched placebo and resolves into a fetch-then-decide circuit—entity-reading heads feeding decision MLPs, predominantly serial by clamping, replicated in architecture on a second family—with dose-response showing the throttle is a matter of magnitude rather than capacity (Section 6).
- **The causal test.** If behavior fails because the network under-reads an adequate signal, conditioning a frozen intervention on the signal itself should restore calibrated behavior with no weight update—which it does, matching a fine-tune while always-on and output-supervised alternatives collapse (Section 7).

2 Related Work

The existence of known/unknown entity features and their causal role is established by Ferrando et al. (2025) and Lindsey et al. (2025); real-time hidden-state probing of hallucinated entities by Obeso et al. (2025); the general internal-versus-expressed dissociation by Orgad et al. (2025); Azaria & Mitchell (2023).

Our framing follows the probing literature’s caveat that a property can be linearly decodable yet not causally used (Ravichander et al., 2021; Hewitt & Liang, 2019; Belinkov, 2022), tested by amnesic interventions (Elazar et al., 2021), and the concurrent argument that verbalized miscalibration is a readout failure remediable by reading an internal signal and steering the output (Miao & Ungar, 2026). We differ from Miao & Ungar (2026) in studying behavioral abstention rather than numeric-confidence calibration, in localizing the causal effect by layer with a norm-matched placebo, and in showing the remedy must be *conditional* rather than always-on. Closest to our thesis, Ruscio & Thompson (2026) independently establish the decodable-but-uncoupled phenomenon: uncertainty is linearly separable in hidden states yet encoded in low-sensitivity subspaces weakly coupled to the unembedding, and a probe-proportional logit bypass restores refusal under random-direction controls. We differ from this work by showing the layer-resolved localization of where the *network itself* reads the signal, the component-level fetch-then-decide circuit, the magnitude account of why natural-scale interventions fail, and the calibration-accounted comparison against weight-level and output-supervised remedies. Latent-knowledge work (Burns et al., 2023; Christiano et al., 2021) and feature case studies (Templeton et al., 2024) are related; Kazemi et al. (2026) independently report an express-versus-represent dissociation (refusal neurons that gate expression versus concept neurons that encode), Xiros et al. (2026) find mathematical-solvability knowledge and its verbalization encoded as distinct linear directions, and Wagner (2026) find hidden-state probes decode question answerability while verbalized answer-confidence is nearly blind to it—the same dissociation in other epistemic domains, without a familiarity construct, localization, or a gated intervention.

Steering by mean-difference or representation engineering (Turner et al., 2024; Li et al., 2023; Zou et al., 2023; Arditi et al., 2024), conditional steering (Lee et al., 2025; Valentino et al., 2026), and the activation-steering/weight-update equivalence (Adila et al., 2026) are the control baselines; Góral et al. (2025) report depth-scheduled steering beating a LoRRA fine-tune on honesty, which is why our claim is about conditionality specifically.

Output-side self-knowledge (Kadavath et al., 2022; Lin et al., 2022; Tian et al., 2023; Xiong et al., 2024), abstention tuning (Zhang et al., 2024), and annotation-efficient honesty alignment via self-consistency (Ni et al., 2026) are the supervision baselines; ReDeEP (Sun et al., 2025) separates parametric and contextual signals in RAG but as complementary covariates of one grounded failure, without a transfer test or an abstain condition.

Relationship to companion manuscripts. A companion applied manuscript analyzes the deployment consequences of the represent-express gap on the same testbed, calibrating operational policy from the rates measured here. The present paper contributes the mechanistic characterization—the familiarity construct, the generalization dissociation, the causal localization of the throttle, and the conditionality result. The intervention comparison of Table 1 is reported in both manuscripts: here as the mechanistic finding, there as

measured technology parameters. A separate developmental study (Nasir, 2026b) traces the training origin of the representation.

Representation-based detectors have been shown to fail out of distribution (Dubanowska et al., 2025). Our concurrent detection paper (Nasir, 2026a) establishes a neuron-level faithfulness detector, causal validation by neuron perturbation (a monotonic dose-response collapse of faithfulness under amplification), the principles *gate rather than correct* and *stratify detectors by mechanism*, and detection gains of the internal signal over output baselines. Conditional steering closes the question that paper leaves open: gating the correction on the internal signal recovers calibrated abstention in place, so gate-rather-than-correct becomes gate-then-correct where the familiarity signal governs.

3 Setup

We construct a testbed of matched parametric pairs (real versus fictional entities) and grounded pairs (answerable versus unsupported questions) whose labels hold by construction, with behavior scored into a three-way categorization by an LLM judge—Claude Sonnet 5 under a fixed model identifier and identical rubric and decode settings across all arms, a model family independent of every subject model and of the item generator: a committed claim, a hedged deflection, or an explicit flag of non-knowledge (a hedged answer that still asserts a specific claim counts as a commit). A stratified 98-item human audit puts label validity at 93.9% (92 of 98), and a rule-based detector agrees with the judged labels at 0.95–1.00 on the arms where both apply. The probe is an ℓ_2 -regularized logistic regression on those representations, evaluated with cross-validation grouped by matched pair. Models are Qwen3-8B (base and instruct), Llama-3.1-8B, Cohere Tiny Aya, and Gemma-3-12B. The probe reads the mean-pooled residual stream at a middle layer (for Llama, the last-token position, its stronger readout). The testbed is built for this research program and is distinct from the SCM-Bench benchmark of the concurrent detection paper (Nasir, 2026a).

Baseline and scale. As a baseline the probe separates real from fictional entities at AUROC 0.992 (95% CI [0.988, 0.997]), is universal across the four families (0.976–0.994; Gemma-3-12B at 0.990), present in the base model, and predicts realized behavior—whether the model in fact commits a fabrication on a given item—at 0.757 [0.731, 0.783], lower than the class separation because behavior varies within each class: unfamiliar entities draw a mix of fabrication, deflection, and flagging that the probe score does not resolve within the class. That behavior is the object we now dissect. The probe is also scale-invariant: replications on Gemma-3-27B-IT, Qwen3-32B, and Qwen3.6-27B under the identical protocol give 0.992–0.994, while the expressed legs of the cascade move independently of capacity—verbalized confidence ranges from near-chance to 0.78 across the three, and clean flagging from 0.056 to 0.642, the high value belonging to the newest-generation Qwen3.6-27B, whose post-training evidently targets abstention yet leaves its verbalized confidence near chance. Capacity fixes neither expressed leg: the training objective moves each separately.

4 Familiarity as Graded Exposure

A probe that separates real from fictional entities could be detecting either existence or familiarity. We separate the two with a five-stratum exposure ladder in which every real entity carries verification evidence gathered during construction: S1, fictional entities (censused for name collisions with real ones); S2, *strictly unseen* real entities, whose first public appearance of any kind—announcement, trailer, leak, or named coverage—postdates the model’s training window (no official knowledge cutoff is documented for Qwen3, but its weights were publicly released on 29 April 2025, so training data necessarily predates that date); S3, *announced-only* entities, named and covered before the window closed but released after it, so they existed in the training data as text about a forthcoming product and never as the product itself; S4, *low-exposure released* entities, real and available well before the window closed but sparsely covered; and S5, well-known anchors. Admission required a stored record per entity (type, earliest documented appearance, release date, source; first public appearance after 1 June 2025, a margin beyond the April release date), and contamination is screened at the level of the exact name string against date-filtered news, forum, and code-hosting corpora, whatever the string’s referent—a search that establishes a strong null rather than proof of absence, with the evidence shipped per entity. The screen evaluated 85 candidate names and rejected 32, leaving 53 entities

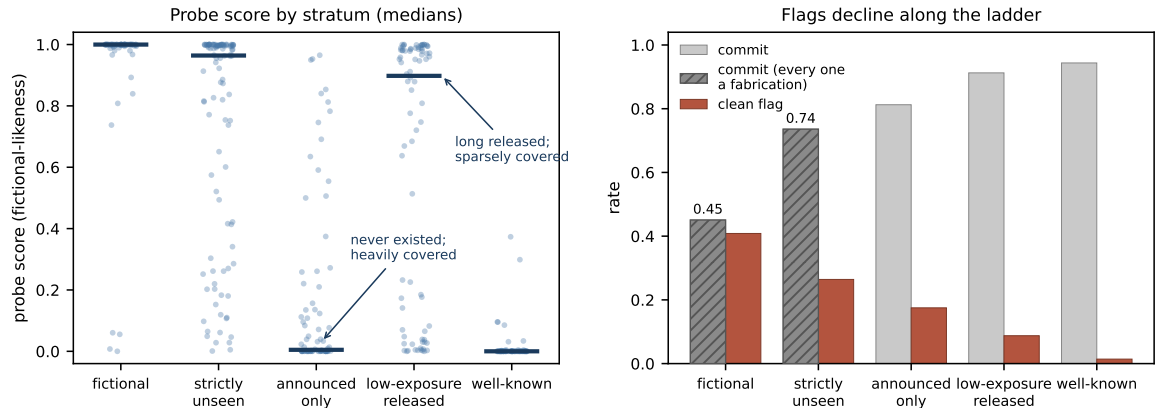


Figure 1: The exposure ladder separates familiarity from existence. Left: per-item probe scores by stratum (horizontal segments mark medians; the values quoted here are stratum means). Announced-only entities, which never existed as products during training, sit near the familiar pole (mean 0.17, median 0.005), while released but sparsely covered entities sit toward the unfamiliar pole (mean 0.62, median 0.90)—the reverse of what an existence detector would produce. Right: commit and clean-flag rates by stratum; flags decline monotonically along the ladder, and in the two fabrication-certain strata (fictional, strictly unseen) every committed answer is a fabrication (0.45 and 0.74).

(106 items). All strata share one question template style. An existence detector orders the strata by whether the entity is real; a familiarity detector orders them by how much text about the entity the model saw.

The ladder orders by exposure, and the diagnostic strata invert existence outright. On a fictional-likeness scale from the probe, the strata score S1 0.933, S2 0.736, S3 0.167, S4 0.623, S5 0.015 (Figure 1). Announced-only entities—products that never existed during training, only text about them—sit near the familiar pole (median 0.005), while released entities with sparse coverage sit toward the unfamiliar pole (median 0.898; Mann–Whitney on entity means, S3 versus S4, $p < 10^{-6}$). An existence detector would order these two strata the other way. The strictly unseen stratum is genuinely intermediate rather than noisy (separation from both poles at $p < 10^{-7}$ on entity means; we test on entity means throughout because items pair within entity), and the wide interquartile ranges within S2 and S4 are informative in the same direction: within a stratum, entities whose names match familiar naming schemas score low and novel-token names score high, and within S2 the heavily covered launches score low while the sparsely covered recent appearances score near the fictional pole, so the signal tracks name and coverage familiarity together, on a continuum for which stratum membership is only a coarse summary.

Behavior follows exposure as well, in two forms. Clean flags (“I don’t recognize this entity”) decline monotonically along the ladder—0.408, 0.264, 0.175, 0.088, 0.014 (Spearman on entity means $\rho = -0.37$, $p < 10^{-9}$)—so the model’s willingness to disclose unfamiliarity follows the designed exposure ranking. At the middle rungs disclosure and the probe diverge: the probe scores announced-only entities as familiar (0.167), yet the model flags them more often than the released stratum (0.175 against 0.088), so behavior follows the design’s exposure ranking at a rung where the probe reads the opposite. Fabrication certainty, however, is confined to the first two strata: only for fictional and strictly-unseen entities is every committed answer necessarily a fabrication, and there the model commits at 0.451 and 0.736 respectively (Fisher exact $p = 1.5 \times 10^{-4}$, item-level). Even in the best case, most questions about verified never-seen real entities receive a confident, specific, and necessarily fabricated answer (78 of 106; exact CI [0.64, 0.82]). Commits in S3 and S4 are not fabrication-certain—the entities exist and the model may hold sparse true knowledge—so we do not read the commit rates as a ladder. One further comparison separates behavior from the probe. Controlling for the probe score itself (an entity-clustered logistic model of committing on probe score and stratum, over all strictly-unseen and low-exposure items), the model commits more readily about sparsely-seen released entities than about never-seen ones at the same probe score (odds ratio 3.4, $p = 0.031$; the same conclu-

sion follows from a Mantel–Haenszel test within probe quintiles, odds ratio 2.8, $p = 0.033$, and a one-sided permutation test on entity means, $p = 0.027$; the three tests triangulate one pre-stated hypothesis rather than confirming independent ones). Descriptively, among items the probe scores above 0.5, low-exposure released entities are committed about at 0.865 against 0.662 for strictly unseen ones at matched probe means (0.93 versus 0.94): behavior distinguishes never-seen from sparsely-seen where the pooled linear probe does not. The behavioral pattern replicates on Llama-3.1-8B, judged under the identical rubric on its own probe scores: the flag gradient holds (entity-level Spearman $\rho = -0.69$, $p < 10^{-36}$; S1 and S2 both near ceiling), and the matched-probe dissociation reappears independently and more strongly (among items above the same probe cut, S4 commits 0.283 against S2’s 0.029; entity-clustered odds ratio 10.4, $p = 0.005$). The levels are model-specific—Llama flags most fabrication-certain items (commits 0.11/0.04) where Qwen answers most—consistent with post-training moving the expressed legs independently (Section 3); the gradient and the dissociation, not the rates, are what transfer.

This refines the standard “known-entity feature” reading: the feature measures how much the model has encountered an entity, rather than whether the entity exists. Concurrent work converges on the same reading by other routes—unsupervised activation-dispersion metrics that track familiarity rather than factual reliability (Brzezinka, 2026a), popularity-stratified familiarity readouts across twelve models, including an unconditional familiarity-direction steer with a monotone refusal response (Brzezinka, 2026b), and evidence that internal states track knowledge recall rather than truthfulness (Cheang et al., 2026)—but none of these designs can separate familiarity from existence, which is what the exposure ladder’s announced-only stratum provides. The distinction also matters for method. Real-versus-fictional contrasts are the standard instrument in this literature, and they cannot separate the two readings: a probe that scores fictional entities high may be detecting either. Post-training-cutoff entities are the standard device for constructing items a model cannot know, and that device assumes anything dated after the cutoff is unfamiliar. The announced-only stratum shows the assumption failing: a product can postdate the window while text about it saturated the training data. The ladder also replicates across families and scales under the identical protocol (stratum means in S1–S5 order): Llama-3.1-8B places the strata at 0.950 / 0.951 / 0.470 / 0.727 / 0.019, Gemma-3-27B at 0.940 / 0.802 / 0.284 / 0.744 / 0.052, Qwen3-32B at 0.904 / 0.805 / 0.226 / 0.659 / 0.043, and Qwen3.6-27B at 0.965 / 0.788 / 0.151 / 0.553 / 0.016—every model reproduces the diagnostic inversion, scoring announced-only entities well below both the fictional and the low-exposure released strata. Sparse presence in the data reads much like absence—the released-but-sparsely-covered stratum sits toward the unfamiliar pole in every family—so familiarity tracks the volume of the model’s exposure to an entity, for which the calendar cutoff is only a proxy. (Qwen3.6 documents no training cutoff, so we do not use its window as evidence; its value here is the protocol-identical replication of the ordering.) Residual undetected contamination can only bias the strictly-unseen stratum toward the familiar pole (a seen entity scores as more familiar), so the clean stratum’s values are conservative bounds. (The rejected candidates themselves score toward the unfamiliar pole: their rejections are mostly name-string collisions with different referents, which the probe, reading the string in context, does not treat as familiar—evidence that the probe is context-sensitive rather than string-matching, and a reason the screen errs on the side of removal.)

5 Two Opposite Generalization Signatures

The model carries two epistemic signals—a parametric knowledge-gap signal (the familiarity signal of Section 4) and a contextual grounding signal (is this answer supported by the passage)—and they are mechanistically distinct: a construct-pure probe trained on one and tested across the boundary transfers near chance (0.50–0.55). Judged-hallucination detectors, which mix mechanisms and surface regularities, show moderate parametric-grounded transfer (0.65–0.77) in concurrent work (Nasir, 2026a), and transfer across languages and domains for most models in a cross-lingual evaluation (Alansari et al., 2026)—consistent with partly distinct machinery rather than full independence; the chance-level result here is specific to probes trained on construct-pure familiarity and groundedness labels. Their *generalization* across distributions is opposite, which is the more informative result.

The knowledge-gap signal is a near-universal circuit. Trained on one domain and tested on unrelated domains—biomedical, public figures, and academic works (121, 128, and 122 matched pairs, against the supply-chain testbed’s 385)—it holds at within-domain AUROC 0.986 and cross-domain AUROC 0.954

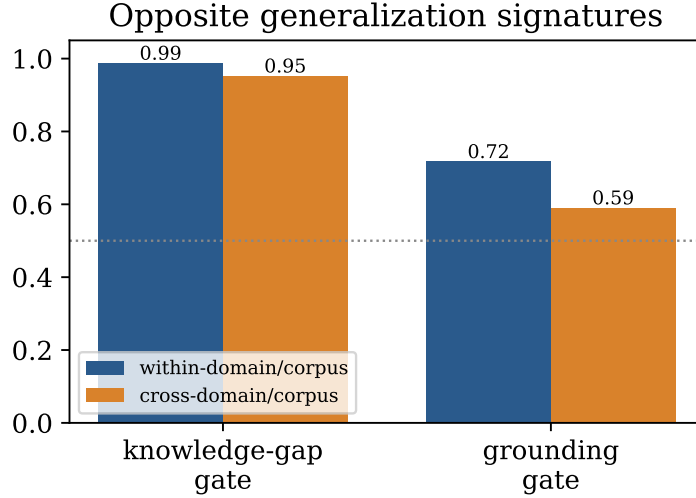


Figure 2: Opposite generalization signatures. The knowledge-gap signal transfers across unrelated domains at 0.95 (a universal circuit); the grounding signal transfers across corpora at only 0.59 (distribution-specific).

(leave-one-domain-out 0.965), a ~ 0.030 drop. The invented entities in the three added domains passed the same name-collision census as the testbed’s fictional tier (Section 4), which dropped 71 colliding pairs; the Llama replication uses an independently run screen. The pattern replicates on Llama-3.1-8B (within 0.984, cross 0.950), and a concurrent study independently finds familiarity probes transferring across entity types at 0.92–0.99 on another family (Brzezinka, 2026a). The grounding probe is distribution-specific. On an independent, human-labeled corpus, RAGTruth (Niu et al., 2024), it is real within-domain (0.717) but transfers across corpora at only 0.59. The specificity attaches to the residual-stream probe of the generator rather than to groundedness detection as such: attention-map features transfer across tasks better than hidden-state classifiers, which reach only 0.56–0.59 in the sliding-window setting (Chuang et al., 2024), and an observer-model grounding direction transfers between same-genre news corpora, while probe performance is lower on an off-genre corpus even when trained in-domain (O’Neill et al., 2025). This is consistent with reports that representation-based hallucination detectors fail out of distribution (Dubanowska et al., 2025) and that hallucination-neuron classifiers do not transfer across domains (Vaddi & Vaddi, 2026); one caveat is that some grounded benchmarks leak label information into prompts (Hussain & Kantarcioglu, 2026), which can distort transfer estimates. Together with the moderate (0.81–0.84) transfer that the concurrent detection paper finds between two structurally similar grounded benchmarks (Nasir, 2026a), the pattern is a distance gradient: grounding transfer decays with distribution distance, while the knowledge-gap signal does not. One signal is portable and the other local (Figure 2); relatedly, Ying et al. (2026) find truth directions ranging from domain-general to domain-specific, with probe geometry predicting transfer—our contribution is the joint measurement—matched probes for the two constructs in one model, cross-construct transfer at chance, and opposite cross-domain signatures—identifying the spectrum’s poles with distinct epistemic constructs. This dissociation—confirming the OOD-failure result for grounding while exhibiting a genuine exception for the knowledge gap—suggests the familiarity circuit is a general property of the pretrained model, whereas groundedness detection rides on distribution-specific surface regularities; that the transfer is not an artifact of a shared item-generation style is supported by the exposure ladder of Section 4, where verified real entities the model has never seen score with the fictional ones.

6 Localizing the Throttle

Layer-resolved readout. Why does a near-perfectly decodable signal reach behavior so weakly? We separate decodability from causal readout, layer by layer, using a norm-matched placebo control. Decodability is saturated from the embeddings onward (AUROC 0.945 at layer 0, ≥ 0.99 by layer 14), so decodability

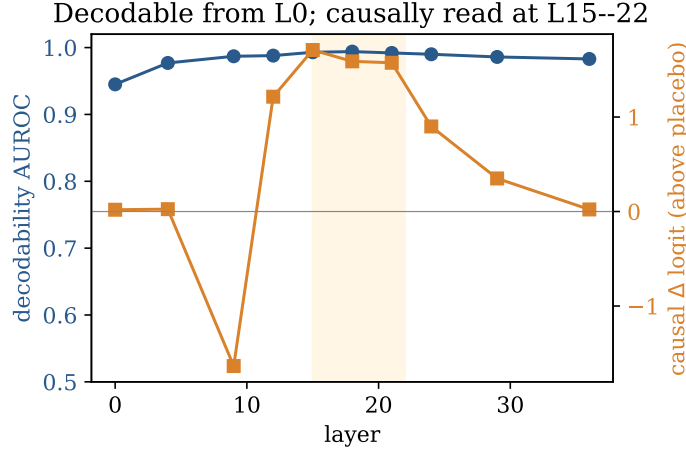


Figure 3: The throttle, localized. Decodability (blue) is saturated from layer 0; the direction-specific causal effect on the abstention decision (orange, above a norm-matched placebo) is concentrated at layers 15–22. The signal is readable everywhere but read into the decision only mid-to-late.

does not localize the readout. The causal readout—how much adding the layer- ℓ mean-difference direction shifts the model toward an abstention continuation, measured against the effect of a random vector of equal norm—is concentrated at layers 15–22 (peak +1.7 logits above placebo at layers 15–21, per-item bootstrap CIs excluding zero), near-zero or negative earlier; the early band is itself anti-abstain (layer 9: −1.6 relative to placebo, CI excluding zero), and the paired difference between bands fixed in advance from an exploratory run (L8–11 versus L14–20) is −1.25, CI [−1.41, −1.08]. The signal is readable everywhere but wired to the abstention decision only in the mid-upper layers (Figure 3).

Component circuit. Component-level patching resolves the readout into a circuit (Figure 4). Screening all 1,188 per-layer attention heads and MLP blocks by gradient attribution (attribution patching; Nanda, 2023; Kramár et al., 2024) and verifying the top candidates by true activation patching on matched real/fictional pairs (the two agree at $r = 0.997$, patching all components reproduces the full representational gap exactly, and every verified component’s bootstrap CI excludes zero against null random controls), the top five components—three decision MLPs (layers 22, 23, 29) and two entity-reading heads (L17.H17, L19.H10)—close 80% of the fictional–real gap and the top twenty close 96%. The largest contributors are MLPs at layers 22–23 (layer 23’s MLP alone closes 41%), joined by attention heads at layers 17–20. Their attention patterns reveal a two-stage mechanism: the early heads attend from the decision position directly to the entity span (97th percentile of attention mass for the strongest), fetching the familiarity signal into the final-position residual, and the late MLPs convert it into the abstain-versus-commit decision; a few heads act as suppressors in the opposite direction. The circuit refines the entity-recognition mechanism of Ferrando et al. (2025)—recognition directions modulating attribute-extraction attention—by exhaustively patching the abstention decision itself, with quantified closure over all components and the decision MLPs identified.

Circuit structure. Two further tests probe its structure. First, the stages are predominantly *serial*: patching the entity-reading heads while clamping the decision MLPs to their fictional-run outputs removes 73% of the heads’ effect (retained fraction 0.27; the collapse is 0.95 in gap-logit units, bootstrap CI [0.80, 1.11], excluding zero), so the heads act predominantly *through* the MLPs; the two heads are mutually near-independent (interaction 0.04) while the decision MLPs overlap substantially (0.29–0.49 pairwise), consistent with redundant copies of one upstream signal. Second, the architecture replicates on Llama-3.1-8B: the same screen-and-verify protocol (1,056 components, full-patch sanity exact) finds entity-reading heads in a mid band (attention mass on the entity span at the 94th–99.9th percentile) feeding late decision MLPs (the layer-31 MLP alone closes 43% of the gap), with model-specific layer positions and lower concentration (top-20 closes 85% against Qwen’s 96%). The fetch-then-decide organization, rather than its exact addresses, is

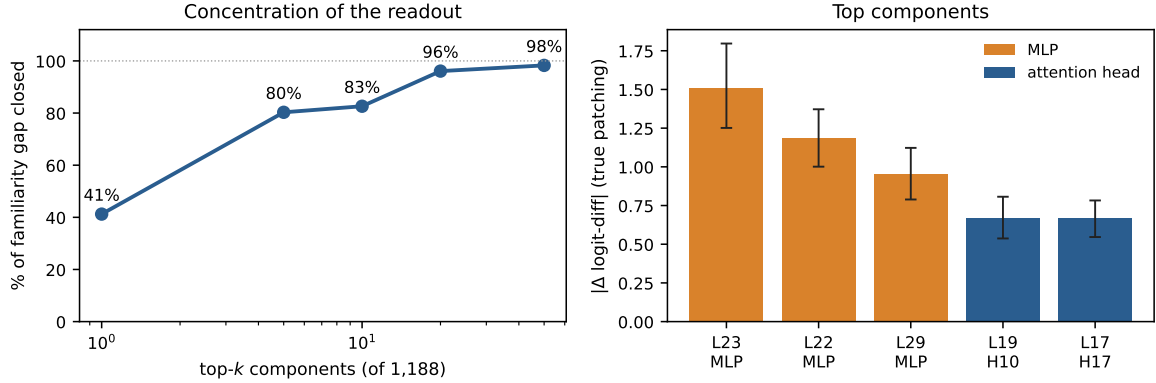


Figure 4: The readout, component-resolved (verified by true activation patching; AtP screen agrees at $r = 0.997$). Left: five of 1,188 components close 80% of the familiarity gap. Right: the largest are decision MLPs at layers 22–23 and entity-reading heads at layers 17–20 (bars: absolute verified effects, 95% CIs).

what transfers. Component contributions concentrate at layers 18–24, slightly later than the steering band, as expected since steering at layer ℓ accumulates all downstream readout. (Caveats: query-head granularity under grouped-query attention; MLP blocks are single coarse components; alignment patches the post-entity suffix, so per-component effects include downstream propagation.)

Dose–response. The throttle is a matter of magnitude rather than capacity, and the dose–response curve of Figure 5 makes this quantitative. Sweeping the optimized direction from $0.1\times$ to $1.25\times$ of its trained norm, clean flagging rises monotonically (0.305, 0.362, 0.430, 0.511, 0.734, 0.905, 0.958) with no threshold discontinuity—the model operates at the low end of a continuous response curve—while false-flags on real entities stay below 0.04 through $0.75\times$, reach 0.189 at the trained norm (a separate decode run of the same condition as the always-on row of Table 1, 0.916/0.182), and collapse to 0.742 at $1.25\times$: calibration sits close below the collapse point, and over-driving the signal reproduces the indiscriminate refusal of output-supervised tuning. Two matched points isolate direction from magnitude. At the natural mean-difference norm ($0.56\times$ the trained norm), the optimized direction already doubles flagging (0.511 against the 0.245 base), while the mean-difference direction at that same norm is behaviorally inert (0.217; concurrent work finds mean-difference steering decisive only when amplified well beyond natural scale, and toward degrading faithfulness rather than inducing abstention (Nasir, 2026a)). The throttle is therefore a property of both direction and magnitude: the behaviorally coupled direction differs from the class-mean axis, and even that direction operates in ordinary forward passes at well below behavioral scale. A monotone refusal response to an unconditional familiarity-direction steer also appears in Brzezinka (2026b), which drives known-entity refusal to 1.00; the calibration axis measured here—false-flags on real entities and the collapse past the trained norm—is what that operating point costs in deployment. So the direction is causally potent—consistent with reports of strong steering effects (Ferrando et al., 2025)—but the model does not deploy it at the magnitude present in ordinary forward passes. The signal is decodable, causally capable, carried by an identifiable fetch-then-decide circuit, and under-used—this combination is the throttle, and it is why reading the signal (a probe) is easy while making the model act on it is not. Concurrent work reaches the same dissociation from other directions: detection directions are largely non-aligned with steering directions across model families (cosine 0.12–0.20) (Galeone et al., 2026), steering along probe-derived directions fails to repair hallucinated *content* across seven models (Roy et al., 2026), and uncertainty has been found to occupy low-sensitivity subspaces weakly coupled to the unembedding, releasable by a probe-proportional logit bypass (Ruscio & Thompson, 2026). The circuit account is the mechanism behind both: the direction is read by a specific, small readout circuit rather than being globally coupled to behavior, and what interventions can recruit is that circuit’s existing output mode, abstention, rather than arbitrary content repair.

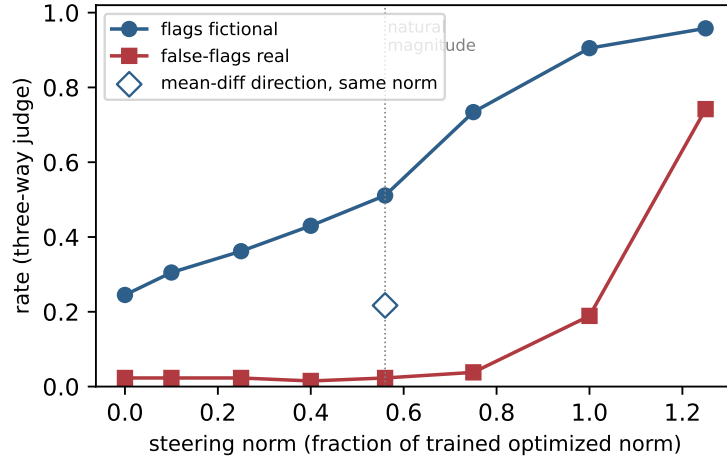


Figure 5: Dose-response of the throttle (held-out split, three-way judge). Flagging rises monotonically with steering magnitude while false-flags stay near zero until the trained norm, then collapse—calibration sits just below a cliff. The open diamond is the mean-difference direction at its natural norm ($0.56\times$): at matched magnitude the optimized direction doubles flagging while the class-mean direction is inert, separating direction quality from magnitude.

method (held-out, three-way judge)	flags fictional	false-flags real	deflects real	flag-gap
base (no intervention)	0.245	0.015	0.553	0.230
mean-difference steer (unconditional)	0.237	0.023	0.545	0.214
optimized steer (unconditional)	0.916	0.182	0.424	0.734
probe-gated steer (CAST, ours)	0.905	0.015	0.523	0.890
probe-tuning (internal-signal fine-tune)	0.895	0.144	0.371	0.751
R-Tuning (output confidence)	0.726	0.674	0.068	0.052
semantic-entropy fine-tune	0.789	0.644	0.129	0.145

Table 1: Releasing the signal (entity-disjoint held-out split, three-way judge, common decode budget). “False-flags real” counts explicit refusals of answerable queries; hedged deflections are tabulated separately, and the flag-gap column (flags fictional minus false-flags real) summarizes calibration. The comparison isolates the gate: an unconditional steer is approximately a fine-tune (Adila et al., 2026), and both trail the conditional arm, so the gated row’s advantage measures the value of conditioning on the internal signal rather than of the intervention itself.

7 Releasing the Signal with Conditional Steering

If the direction is potent but under-used, the control question is how to deploy it without breaking calibration. Table 1 compares interventions on an entity-disjoint held-out split, scored by the three-way judge at a common decode budget; across them, conditionality—gating the intervention on the signal itself—is the property that separates calibrated from indiscriminate abstention.

The optimized vector is a single additive direction at layer 18, learned by gradient descent through the frozen model to favor an abstention continuation on fictional training entities and the model’s own answer on real ones, with training entities disjoint from evaluation; the output-supervised baselines fine-tune the same adapter architecture on pseudo-labels from output confidence and from ten-sample semantic entropy (Farquhar et al., 2024), following the semantic-entropy abstention-tuning recipe of Tjandra et al. (2024). Applied always-on, the optimized vector forces flagging on fictional entities (0.916) but explicitly false-flags real ones at 0.182; a mean-difference steer at natural scale fails to induce abstention (0.237 against the

0.245 untouched base on the held-out arm; the dose-response arm of Section 6 measures the same inertness at 0.217). Applying the same vector *only when the probe fires*—conditional activation steering (Lee et al., 2025)—holds flagging at 0.905 while false-flagging real entities at 0.015, the untouched model’s own rate, with hedged deflections essentially unchanged—below both the unconditional steer’s 0.182 and the probe-supervised low-rank fine-tune’s 0.144 (non-overlapping exact binomial intervals)—at inference time with no weight training. Summarized as the flag-gap (flagging on fictional minus false-flagging on real), the gated vector leads every arm: 0.890, against 0.751 for the fine-tune and 0.734 for the always-on steer. Given the first-order equivalence between activation steering and weight updates (Adila et al., 2026), this is the expected result: since an unconditional steer is approximately a fine-tune, the win must come from the gate.

It also instantiates the classifier-gated correction scheme the concurrent detection paper leaves open when reporting that unconditional steering cannot repair grounded faithfulness (Nasir, 2026a): abstention is an existing mode that can be induced; faithful content cannot. Gating an intervention on a detector is also explored by Nielsen et al. (2026), whose probabilistic-circuit gate triggers contrastive *decoding* to repair content; our remedy differs in operating at the activation level, targeting abstention rather than content repair, and carrying fine-tuning comparisons. The closed-form error-reduction steering of Hedström et al. (2025), which abstains from intervening when no reliable correction exists, is the nearest control-side neighbor; our delta is the familiarity-probe gate and the head-to-head against weight-level and output-supervised remedies.

In a live two-step agent—the model extracts a value from a document and then computes with it; a fabricated extraction that survives into the final answer is a *cascading* error, and *escalation* routes an item to a human instead of answering—the same probe-gated vector converts fabricated extractions into well-formed in-place abstentions on parametric turns ($0.833 \rightarrow 0.082$ cascading, utility unchanged, zero escalations). On grounded turns it fails twice over: the entity-trained gate’s scores saturate on long agentic prompts and cease to discriminate (the signal itself survives the format shift at 0.986 AUROC; the failure is per-format threshold calibration), and even a domain-matched, grounded-trained gate—which fires correctly—reduces cascading only modestly ($0.139 \rightarrow 0.111$), because the vector loses exactly on confidently committed fabrications and occasionally induces one. Conditional steering is only as portable as its trigger, and beyond that, only as strong as the behavior it must overcome: in-place correction works where the familiarity signal governs (parametric recall) and fails where confident grounded fabrication does. This qualifies recent reports that steering can outperform a fine-tuning baseline for honesty (Góral et al., 2025): the operative ingredient is conditionality tied to the internal signal.

The scheme also survives production serving conditions. In a separate systems implementation (manuscript in preparation), the probe and the gated vector run inside a serving engine’s continuously batched forward pass—scored per token in-graph, gated per request, steered through a fused kernel—with no measurable throughput cost over the engine’s optimized baseline. In-engine, judged by the same three-way judge on matched items, the gated intervention cuts committed fabrications from 0.388 to 0.212 and raises clean flagging from 0.197 to 0.679 (the gate firing on 83% of fictional and 6% of real requests) while leaving the real side unchanged (0.010 false-flags), and mixed-batch outputs are identical to single-request execution. The mechanism does not depend on the research harness.

The conditionality effect replicates on Llama-3.1-8B: the unconditional optimized vector there collapses calibration (flagging 0.958 of fictional while false-flagging 0.894 of real, a flag-gap of 0.064 against the untouched baseline’s 0.212), while gating the same vector on that model’s own familiarity probe yields the best flag-gap of any arm (0.263; false-flags cut from 0.894 to 0.674 at near-equal flagging). The absolute calibration ceiling differs, however, because Llama blanket-refuses at baseline (0.682 false-flags on real entities with no intervention): conditional steering prevents the over-refusal the intervention would add, but cannot remove what the base model does on its own.

Supervision from the output side cannot substitute. R-Tuning-style tuning (Zhang et al., 2024) and semantic-entropy tuning (Tjandra et al., 2024) both collapse to indiscriminate refusal (explicit false-flag rates of 0.674 and 0.644 on real entities), because answer-level output confidence cannot distinguish confident fabrication from correct recall: the model’s mean log-probability is -0.20 on fictional versus -0.21 on real entities, and semantic entropy separates them at only 0.546 AUROC. This closes the counterargument that a richer output signal (self-consistency or semantic entropy) would rescue confidence-based remediation (Ni et al.,

2026)—in the familiarity setting it does not. Calibrated abstention is learnable, but the supervision must come from the internal familiarity signal, and the deployment must be conditional on it.

8 Discussion

Three properties organize the represent-express gap. The signal is *familiarity*: the model represents acquaintance rather than existence, and so does not know it does not know a genuinely new entity. It has *opposite generalization signatures*: the knowledge-gap circuit is universal across domains while groundedness is distribution-specific. And it is *throttled*: decodable from the embeddings, causally wired to the decision only at layers 15–22—by a verified fetch-then-decide circuit of entity-reading heads and decision MLPs—and causally potent but under-used at natural magnitude, so it is released by conditionality, with or without weight updates. Together these say the failure to abstain is under-driven wiring rather than missing knowledge—the circuit is present but operates below behavioral magnitude—from a familiarity representation to an abstention action. This is congruent with the global-workspace account of verbalizable representations (Gurnee et al., 2026): models maintain far more internal structure than their output policy consumes, and the operative question—which our companion developmental study takes up (Nasir, 2026b)—is what training determines which signals the policy reads.

8.1 Limitations

Each probe is per-model. The exposure ladder’s probe ordering replicates on four further models across three families (Llama-3.1-8B, Gemma-3-27B, Qwen3-32B, Qwen3.6-27B), and its behavioral pattern—the flag gradient and the matched-probe dissociation—replicates on Llama-3.1-8B, though behavioral rates are model-specific and the remaining three models have probe-side replication only; the knowledge-gap transfer and the conditionality effect replicate on a second family (Llama-3.1-8B); the grounding-transfer measurement and the layer-resolved causal band are single-model (Qwen3-8B); the component-level circuit replicates in architecture on Llama-3.1-8B, with model-specific layer bands and lower concentration. Models span 3–32B parameters for the representational, familiarity, and behavioral results. The causal localization uses a mean-difference direction and a fixed abstention-versus-commit metric; the component analysis is verified by true activation patching but at query-head granularity under grouped-query attention and with suffix-only alignment. The exposure ladder’s dating is bounded by the model’s public release date rather than a documented cutoff, and its verification searches—date-filtered news, forum, and code-hosting corpora—establish a strong null rather than proof of absence; since residual undetected contamination could only add exposure and push scores toward the familiar pole, the strictly-unseen stratum’s values are conservative bounds. Behavioral labels come from a single primary LLM judge (Claude Sonnet 5, fixed model identifier, a model family independent of every subject model), and the testbed’s construction labels are validated by a stratified 98-item human audit (93.9% label validity: 92 of 98). The judge’s verdicts are cross-checked three ways: a rule-based detector (0.95–1.00 agreement on the arms where both apply), an independent open-weights judge (Qwen3-32B, 0.84), and a full re-judge of the corpus, every exposure-ladder arm included, by a second frontier judge from an independent family (GPT-5.6; 11,445 paired verdicts, agreement 0.751, $\kappa = 0.593$ overall; 0.89–0.99 across the Qwen ladder arms and 0.74 on the Llama ladder arm, whose refusals-with-residual-content straddle the flag/deflect boundary; disagreements concentrated in adjacent categories, and every ordering the paper reports—including the full ladder and the matched-probe direction—reproducing under the alternative judge); a verdict-level human audit of the judge itself (blind, 150 stratified responses) agrees at 0.720 ($\kappa = 0.574$), with 3 of 150 disagreements skipping a category and the judge shifting human commits and flags into deflections—so the commit and flag rates reported here are conservative. Data are English-only.

Reproducibility Statement

A replication package accompanies the paper (<https://github.com/murtaza-nasir/familiarity-throttle>). It contains the exposure-ladder dataset with per-entity verification and rejection evidence, all generations and both judge families’ verdicts for every arm (Sections 4 and 7), the

cross-domain transfer data and scripts (Section 5), the Section 6 result files (layer-resolved causal trace, component screen and verification, circuit dependency, dose-response), the judge triangulation record, and the analysis and figure scripts that reproduce every reported number. Raw activation caches are too large to ship and remain on the compute pipeline; their generating scripts are included.

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